

PRACTICAL ACTIVITY 2: Performing Frequency Analysis to Partially Break a Substitution Cipher

Objective:

The objective of this activity is to apply frequency analysis to partially break a monoalphabetic substitution cipher. By examining how often each letter appears in the ciphertext and comparing these frequencies with typical English letter distributions, it is possible to make educated guesses about the plaintext and gradually reveal the hidden message.

CIPHERTEXT TO SOLVE:

“
GFS WMY OG LGDVS MF SFNKYHOSU ESLLMRS, PC WS BFGW POL DMFROMRS, PL OG CPFU M UPCCSKSFO HDMPFOSXO GC OIS LMES
DMFROMRS DGFR SFGQRI OG CPDD GFS LISSO GK LG, MFU OISF WS NGQFO OIS GNNQKKSFNSL GC SMNI DSOOSK. WS NMDD OIS EGLO
CKSJQSFOY GNNQKKPFR DSOOSK OIS 'CPKLO'; OIS FSXO EGLO GNNQKKPFR DSOOSK OIS 'LSNGFU' OIS CGDDGWPFR EGLO GNNQKKPFR
DSOOSK OIS 'OIPKU', MFU LG GF, QFOPD WS MNNGQFO CGK MDD OIS UPCCSKSFO DSOOSK PF OIS HDMPFOSXO LMEHDS. OISF WS DGGB MO
OIS NPHISK OSXO WS WMFO OG LGDVS MFU WS MDLG NDMLLPCY POL LYEAGDL WS CPFU OIS EGLO GNNQKKPFR LYEAGD MFU NIMFRS PO OG
OIS CGKE GC OIS 'CPKLO' DSOOSK GC OIS HDMPFOSXO LMEHDS, OIS FSXO EGLO NGEEGF LYEAGD PL NIMFRSU OG OIS CGKE GC OIS 'LSNGFU'
DSOOSK, MFU OIS CGDDGWPFR EGLO NGEEGF LYEAGD PL NIMFRSU OG OIS CGKE GC OIS 'OIPKU' DSOOSK, MFU LG GF, QFOPD WS MNNGQFO
CGK MDD LYEAGDL GC OIS NKYHOGRKME WS WMFO OG LGDVS.

Decryption Process:

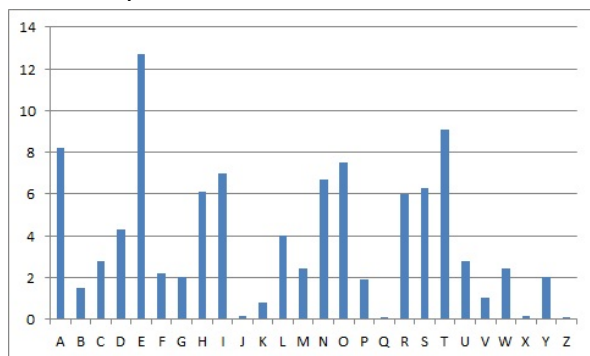
The first step in frequency analysis is to count the number of times each letter appears in the ciphertext. This produces a frequency distribution of the encrypted letters.

Ciphertext Letter	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	5	2	26	42	23	51	67	8	33	1	35	39	35	29	85	30	14	17	88	0	17	3	16	6	10	0

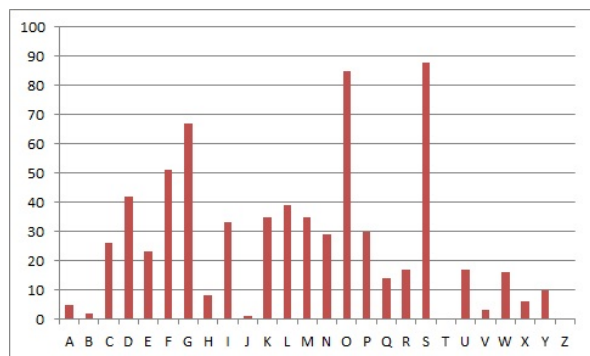
If ordered from most common to least to make comparisons easier:

Ciphertext Letter	S	O	G	F	D	L	K	M	I	P	N	C	E	R	U	W	Q	Y	H	X	A	V	B	J	T	Z
Frequency	88	85	67	51	42	39	35	35	33	30	29	26	23	17	17	16	14	10	8	6	5	3	2	1	0	0

As a comparison:



The Standard English Letter Frequencies



The letter frequencies of the letters in the ciphertext

By comparing the ciphertext distribution with these known patterns, possible substitutions can be inferred.

Initial Substitution Guess:

Since E is the most common letter in English, the most frequently occurring letter in the ciphertext is likely to represent E (*From the ciphertext analysis: S → E*). Another frequent letter in the ciphertext is O, which may represent T (*O → T*). **After applying these substitutions, some recognizable patterns begin to appear in the encrypted text.**

GFe WMY tG LGDVe MF eFNKYHteU EeLLMRe, PC We BFGW PtL
DMFRQMRe, PL tG CPFU M UPCCeKeFt HDMPFteXt GC tle LMEe
DMFRQMRe DGFR eFGQRI tG CPDD GFe Lleet GK LG, MFU tleF We NGQFt
tle GNNQKKeFNeL GC eMNI DetteK. We NMDD tle EGLt CKeJQeFtDY
GNNQKKPFR DetteK tle 'CPKLt', tle FeXt EGLt GNNQKKPFR DetteK tle
'LeNGFU' tle CGDDGWPFR EGLt GNNQKKPFR DetteK tle 'tIPKU', MFU LG
GF, QFtPD We MNNGQFt CGK MDD tle UPCCeKeFt DetteKL PF tle
HDMPFteXt LMEHDe. tleF We DGGB Mt tle NPHleK teXt We WMFt tG LGDVe
MFU We MDLG NDMLPCY PtL LYEAGDL. We CPFU tle EGLt GNNQKKPFR
LYEAGD MFU NIMFRe Pt tG tle CGKE GC tle 'CPKLt' DetteK GC tle
HDMPFteXt LMEHDe, tle FeXt EGLt NGEEGF LYEAGD PL NIMFReU tG tle
CGKE GC tle 'LeNGFU' DetteK, MFU tle CGDDGWPFR EGLt NGEEGF
LYEAGD PL NIMFReU tG tle CGKE GC tle 'tIPKU' DetteK, MFU LG GF, QFtPD
We MNNGQFt CGK MDD LYEAGDL GC tle NKYHtGRKME We WMFt tG
LGDVe.

Identifying Common Word Patterns:

After applying the initial substitutions, certain short word patterns begin to resemble common English words. One repeated pattern appears similar to: **tle**. In English, the most common three-letter word is **"the."** From this observation, we can deduce: **I** → **H**. Once this substitution is applied, the word becomes: **the**. Also, by looking at the frequencies again, we see the next most common letter is **"G"**, which is probably one of **"a"**, **"i"** or **"o"**. We see that the third word is **"tG"**, and the only one of these options that makes sense is **"to"**, so we guess **G** → **o**.

oFe WMY to LoDVe MF eFNKYHteU EeLLMRe, PC We BFoW PtL DMFRQMRe, PL to CPFU
M UPCCeKeFt HDMPFteXt oC the LMEe DMFRQMRe DoFR eFoQRh to CPDD oFe Lheet oK
Lo, MFU theF We NoQft the oNNQKKeFNeL oC eMNH DetteK. We NMDD the EoLt
CKeJQeFtDY oNNQKKPFR DetteK the 'CPKLt', the FeXt EoLt oNNQKKPFR DetteK the
'LeNoFU' the CoDDoWPFR EoLt oNNQKKPFR DetteK the 'thPKU', MFU Lo oF, QFtPD We
MNNoQft CoK MDD the UPCCeKeFt DetteKL PF the HDMPFteXt LMEHDe. theF We DooB
Mt the NPHheK teXt We WMft to LoDVe MFU We MDLo NDMLLPCY PtL LYEAoDL. We
CPFU the EoLt oNNQKKPFR LYEAoD MFU NhMFRe Pt to the CoKE oC the 'CPKLt' DetteK
oC the HDMPFteXt LMEHDe, the FeXt EoLt NoEEoF LYEAoD PL NhMFReU to the CoKE oC
the 'LeNoFU' DetteK, MFU the CoDDoWPFR EoLt NoEEoF LYEAoD PL NhMFReU to the
CoKE oC the 'thPKU' DetteK, MFU Lo oF, QFtPD We MNNoQft CoK MDD LYEAoDL oC the
NKYHtoRKME We WMft to LoDVe.

The first word is now **"oFe"**, which when considered with the appearance of **"theF"**, leads us to the conclusion that **F** → **n**. This also fits in with the frequencies of both letters in the tables. In the third line we see the word **"Lheet"**, which is most likely to be **"sheet"**, and so we replace **L** → **s**. Again, the frequencies of these two letters are about right.

one WMY to soDVe Mn enNKYHteU EessMRe, PC We BnoW Pts DMnRQMRe, Ps to CPnU M
UPCCeKent HDMPnteXt oC the sMEe DMnRQMRe DonR enoQRh to CPDD one sheet oK
so, MnU then We NoQnt the oNNQKKenNes oC eMNH DetteK. We NMDD the Eost
CKeJQentDY oNNQKKPnR DetteK the 'CPKst', the neXt Eost oNNQKKPnR DetteK the
'seNonU' the CoDDoWPnR Eost oNNQKKPnR DetteK the 'thPKU', MnU so on, QntPD We
MNNoQnt CoK MDD the UPCCeKent DetteKs Pn the HDMPnteXt sMEHDe. then We DooB
Mt the NPHheK teXt We WMnt to soDVe MnU We MDso NDMssPCY Pts sYEAoDs. We CPnU
the Eost oNNQKKPnR sYEAoD MnU NhMnRe Pt to the CoKE oC the 'CPKst' DetteK oC the
HDMPnteXt sMEHDe, the neXt Eost NoEEon sYEAoD Ps NhMnReU to the CoKE oC the
'seNonU' DetteK, MnU the CoDDoWPnR Eost NoEEon sYEAoD Ps NhMnReU to the CoKE
oC the 'thPKU' DetteK, MnU so on, QntPD We MNNoQnt CoK MDD sYEAoDs oC the
NKYHtoRKME We WMnt to soDVe.

We see the word **"soDVe"**, which could be **"solve"**, implying the transformations of **"D"** and **"V"** to **"I"** and **"v"** respectively. In the second line we now have the phrase **"one sheet oK so"**, which suggests that **K → r**.

one WMY to solve Mn enNrYHteU EessMRe, PC We BnoW Pts IMnRQMRe, Ps to CPnU M UPCCerent HIMPnteXt oC the sMEe IMnRQMRe lonR enoQRh to CPII one sheet or so, MnU then We NoQnt the oNNQrrrenNes oC eMNH letter. We NMII the Eost CreJQentlY oNNQrrPnR letter the 'CPrst', the neXt Eost oNNQrrPnR letter the 'seNonU' the ColloWPnR Eost oNNQrrPnR letter the 'thPrU', MnU so on, QntPI We MNNoQnt Cor MII the UPCCerent letters Pn the HIMPnteXt sMEHle. then We looB Mt the NPHher teXt We WMnt to solve MnU We MIso NIMssPCY Pts sYEAols. We CPnU the Eost oNNQrrPnR sYEAol MnU NhMnRe Pt to the CorE oC the 'CPrst' letter oC the HIMPnteXt sMEHle, the neXt Eost NoEEon sYEAol Ps NhMnReU to the CorE oC the 'seNonU' letter, MnU the ColloWPnR Eost NoEEon sYEAol Ps NhMnReU to the CorE oC the 'thPrU' letter, MnU so on, QntPI We MNNoQnt Cor MII sYEAols oC the NrYHtoRrME We WMnt to solve.

In the middle of the second line we have the word **"enoQRh"**, which is likely to be **"enough"**, and so we have the transformations **"Q"** and **"R"** to **"u"** and **"g"** respectively.

one WMY to solve Mn enNrYHteU EessMge, PC We BnoW Pts IMnguMge, Ps to CPnU M UPCCerent HIMPnteXt oC the sMEe IMnguMge long enough to CPII one sheet or so, MnU then We Nount the oNNurrenNes oC eMNH letter. We NMII the Eost CreJuentlY oNNurrPng letter the 'CPrst', the neXt Eost oNNurrPng letter the 'seNonU' the ColloWPng Eost oNNurrPng letter the 'thPrU', MnU so on, untPI We MNNount Cor MII the UPCCerent letters Pn the HIMPnteXt sMEHle. then We looB Mt the NPHher teXt We WMnt to solve MnU We MIso NIMssPCY Pts sYEAols. We CPnU the Eost oNNurrPng sYEAol MnU NhMnge Pt to the CorE oC the 'CPrst' letter oC the HIMPnteXt sMEHle, the neXt Eost NoEEon sYEAol Ps NhMngeU to the CorE oC the 'seNonU' letter, MnU the ColloWPng Eost NoEEon sYEAol Ps NhMngeU to the CorE oC the 'thPrU' letter, MnU so on, untPI We MNNount Cor MII sYEAols oC the NrYHtogrME We WMnt to solve.

We have the word **"Nount"** which is could be **"count"** and **"EessMge"** which is likely to be **"message"**, giving us that **N → c**, **E → m** and **M → a**.

one WaY to solve an encrYHteU message, PC We BnoW Pts language, Ps to CPnU a UPCCerent HlaPnteXt oC the same language long enough to CPII one sheet or so, anU then We count the occurrences oC each letter. We call the most CreJuentlY occurPng letter the 'CPrst', the neXt most occurPng letter the 'seconU' the ColloWPng most occurPng letter the 'thPrU', anU so on, untPI We account Cor all the UPCCerent letters Pn the HlaPnteXt samHle. then We looB at the cPHher teXt We Want to solve anU We also classPCY Pts sYmAols. We CPnU the most occurPng sYmAol anU change Pt to the Corm oC the 'CPrst' letter oC the HlaPnteXt samHle, the neXt most common sYmAol Ps changeU to the Corm oC the 'seconU' letter, anU the ColloWPng most common sYmAol Ps changeU to the Corm oC the 'thPrU' letter, anU so on, untPI We account Cor all sYmAols oC the crYHtogram We Want to solve.

It is likely that $W \rightarrow w$, $X \rightarrow x$, $Y \rightarrow y$ and $Z \rightarrow z$. The word "occurrPng" is clearly meant to read "occurring", and it is likely that "sYmAol" is "symbol".

one way to solve an encryHteU message, iC we Bnow its language, is to CinU a UiCCerent Hlaintext oC the same language long enough to Cill one sheet or so, anU then we count the occurrences oC each letter. we call the most CreJuently occurring letter the 'Cirst', the next most occurring letter the 'seconU' the Collowing most occurring letter the 'thirU', anU so on, until we account Cor all the UiCCerent letters in the Hlaintext samHle. then we looB at the ciHher text we want to solve anU we also classiCy its symbols. we CinU the most occurring symbol anU change it to the Corm oC the 'Cirst' letter oC the Hlaintext samHle, the next most common symbol is changeU to the Corm oC the 'seconU' letter, anU the Collowing most common symbol is changeU to the Corm oC the 'thirU' letter, anU so on, until we account Cor all symbols oC the cryHtogram we want to solve.

We can now see that $C \rightarrow f$, $B \rightarrow k$, $U \rightarrow d$, $J \rightarrow q$ and $H \rightarrow p$.

Decrypted Message:

one way to solve an encrypted message, if we know its language, is to find a different plaintext of the same language long enough to fill one sheet or so, and then we count the occurrences of each letter. we call the most frequently occurring letter the 'first', the next most occurring letter the 'second' the following most occurring letter the 'third', and so on, until we account for all the different letters in the plaintext sample. then we look at the cipher text we want to solve and we also classify its symbols. we find the most occurring symbol and change it to the form of the 'first' letter of the plaintext sample, the next most common symbol is changed to the form of the 'second' letter, and the following most common symbol is changed to the form of the 'third' letter, and so on, until we account for all symbols of the cryptogram we want to solve.

Reflection:

This activity provided a deeper understanding of how classical cryptographic systems can be analyzed and broken using statistical methods. By applying frequency analysis to the given substitution cipher, I learned how patterns in language can reveal hidden information even when the original message has been encrypted. Although the ciphertext initially appeared completely random, examining the frequency of letters and comparing them with the typical frequency distribution in English allowed me to make logical guesses about possible substitutions. Gradually, recognizable word patterns such as common three-letter words helped confirm these guesses and made it possible to reconstruct portions of the original message.

Through this process, I realized that substitution ciphers rely on replacing letters consistently, which preserves the statistical structure of the language. Because of this, attackers can exploit these patterns to break the encryption without knowing the key. The exercise demonstrated that even simple analytical techniques can reveal weaknesses in classical cryptographic systems. It also highlighted how cryptanalysis often involves a combination of mathematical reasoning and linguistic intuition, such as recognizing common words, letter pairs, and sentence structures.

Overall, this activity reinforced the importance of stronger encryption methods in modern cybersecurity. While substitution ciphers played an important role in the historical development of cryptography, they are no longer secure against modern analysis techniques. Understanding how these ciphers can be broken provides valuable insight into why modern encryption algorithms must eliminate predictable patterns and rely on more complex mathematical structures to ensure data security.